Move Element To End

AlgoExpert · Medium · Arrays

The problem

- In place, move every occurrence of `toMove` to the end (other order irrelevant).

Intuition

- Two pointers: move the target from the left end out of the way by swapping with non-targets from the right.

Approach

- 1. Left and right pointers
- 2. Skip targets already at the right
- 3. Swap a target on the left with a non-target on the right
- 4. Meet in the middle

Java solution

```java
public class MoveElementToEnd {
    public static List<Integer> moveElementToEnd(List<Integer> array, int toMove) {
        int l = 0, r = array.size() - 1;
        while (l < r) {
            while (l < r && array.get(r) == toMove) r--;
            if (array.get(l) == toMove) Collections.swap(array, l, r);
            l++;
        }
        return array;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Two pointers: move the target from the left end out of the way by swapping with non-targets from the right.
- Complexity: O(n)